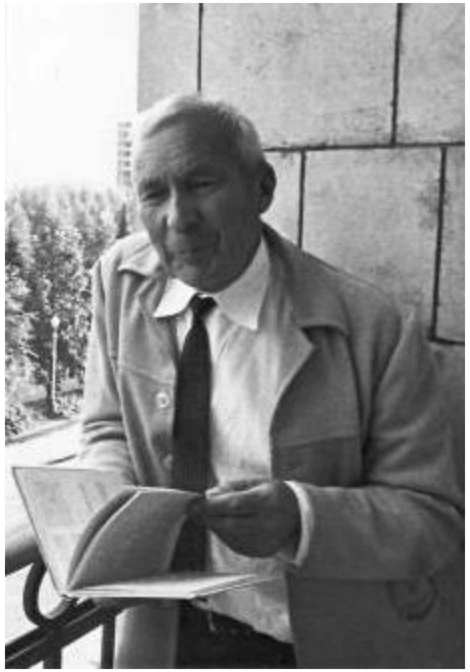
Math Primer
(18. Conditional Expectation)

Contents



A. Kolmogorov
(1903-1907)

Conditional Expectation (Motivation)

- Conditional Expectation is the backbone of derivative pricing and many other use cases
- The price of a derivative security at time t is defined as expected discounted payoff conditional on filtration at t i.e., information available till t.

$$P(t,T) = \mathbb{E}\left[\exp\left(-\int_t^T r(s)ds\right) \mid \mathcal{F}_t\right]$$

- Any prediction or forecasting exercise involves calculation of conditional expectation y given x

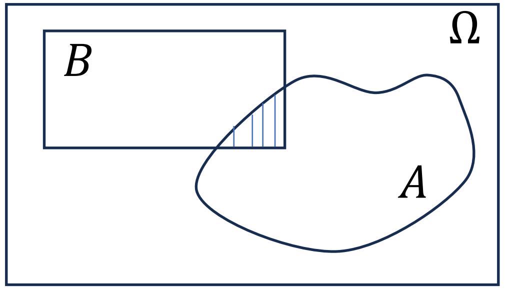
$$\mathbb{E}[y \mid X] = X\beta$$

- In cases where any convolution of two random variables is involved, we use the property of iterated expectations. (more later.)

Conditional Expectation

- Conditional Expectation is calculated using the conditional probabilities

(Conditional Probability)



$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

(Discrete Random)

$$\mathbb{E}[X] = \sum_x x\mathbb{P}\,(X = x)$$

$$\mathbb{E}[X|Y = y] = \sum_x x\ \mathbb{P}(X = x\ |Y = y)$$

$$= \sum_x x\ \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)}$$

(Continuous Random)

$$\mathbb{E}[X] = \int_\Omega x f_X(x) dx$$

$$\mathbb{E}[X|Y = y] = \int_\Omega x f_{X|Y}(x|y) dx$$

$$= \int_\Omega x\ \frac{f_{X,Y}(x,y)}{f_Y(y)} dx$$

Conditional Expectation

PEAKS2TAILS

Exercise 1 - Consider the roll of a die. Let $X = 1$ if number is even and 0 otherwise. Let $Y = 1$ if the number is prime. Find (I) $\mathbb{E}[X]$, (II) $\mathbb{E}[X|Y = 1]$

ω	1	2	3	4	5	6
$\mathbb{P}(\omega)$	1/6	1/6	1/6	1/6	1/6	1/6
X	0	1	0	1	0	1
Y	0	1	1	0	1	0

Conditional Expectation

Exercise 2 – The joint density of 2 random variables is given as

$$f_{X,Y}(x,y) = x^2 + \frac{4y}{3} \qquad 0 \le x \le 1\,, 0 \le y \le 1$$

Show that this is a proper density and find $\mathbb{E}[X|Y=y]$

Conditional Expectation as Random Variable

- Consider a function $h(x) = x^2$. If X is a random variable that takes value x, what is $h(X)$?
- $h(X)$ is a random variable that takes a value x^2 when X takes a value x. In general, a function of a random variable is another random variable. Continuing this thought, the conditional expectation $\mathbb{E}[X|Y = y]$ is a function of y. However, if we do not specify a fixed value y, then the quantity $\mathbb{E}[X|Y]$ is a random variable which basically will take random values as Y takes random values.
- As any random variable $\mathbb{E}[X|Y]$ will have distribution, expectation, variance etc....

$$g(y) = \mathbb{E}[X|Y = y]$$

$$g(Y) = \mathbb{E}[X|Y]$$ $g(Y)$ is a random variable

Law of total expectation

- The mean of $\mathbb{E}[X|Y]$ is referred to as law of total expectation or law of iterative expectation or Tower rule. It's defined as

$$\mathbb{E}\big[\mathbb{E}[X|Y]\big] = \mathbb{E}[X]$$

Informally speaking, the expectation of conditional expectation is the unconditional expectation. Here is the proof assuming discrete random

$$\mathbb{E}\big[\mathbb{E}[X|Y]\big] = \sum_y \mathbb{E}[X|Y=y]\mathbb{P}(Y=y)$$

$$\implies \mathbb{E}\big[\mathbb{E}[X|Y]\big] = \sum_y \sum_x x \frac{\mathbb{P}(X=x, Y=y)}{\mathbb{P}(Y=y)} \mathbb{P}(Y=y)$$

$$\implies \mathbb{E}\big[\mathbb{E}[X|Y]\big] = \sum_x x\mathbb{P}(X=x) = \mathbb{E}[X]$$

Law of total expectation

Exercise 3 – In a school, there are two sections for grade 10 ;

section $y = 1$(10 students) and section $y = 2$(20 students) , x_i is the score of a student

Data : $y = 1$: $\frac{1}{10}\sum_{i=1}^{10} x_i = 90$ $y = 2$: $\frac{1}{20}\sum_{i=11}^{30} x_i = 60$

Find $\mathbb{E}[X]$, $\mathbb{E}[X|Y = 1]$, $\mathbb{E}[X|Y = 2]$. Verify the Tower property.

Conditional Variance as a Random Variable

- Similar to the idea of a conditional expectation, conditional variance is a random variable

$$\text{var}\,(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] \qquad \text{(unconditional variance)}$$

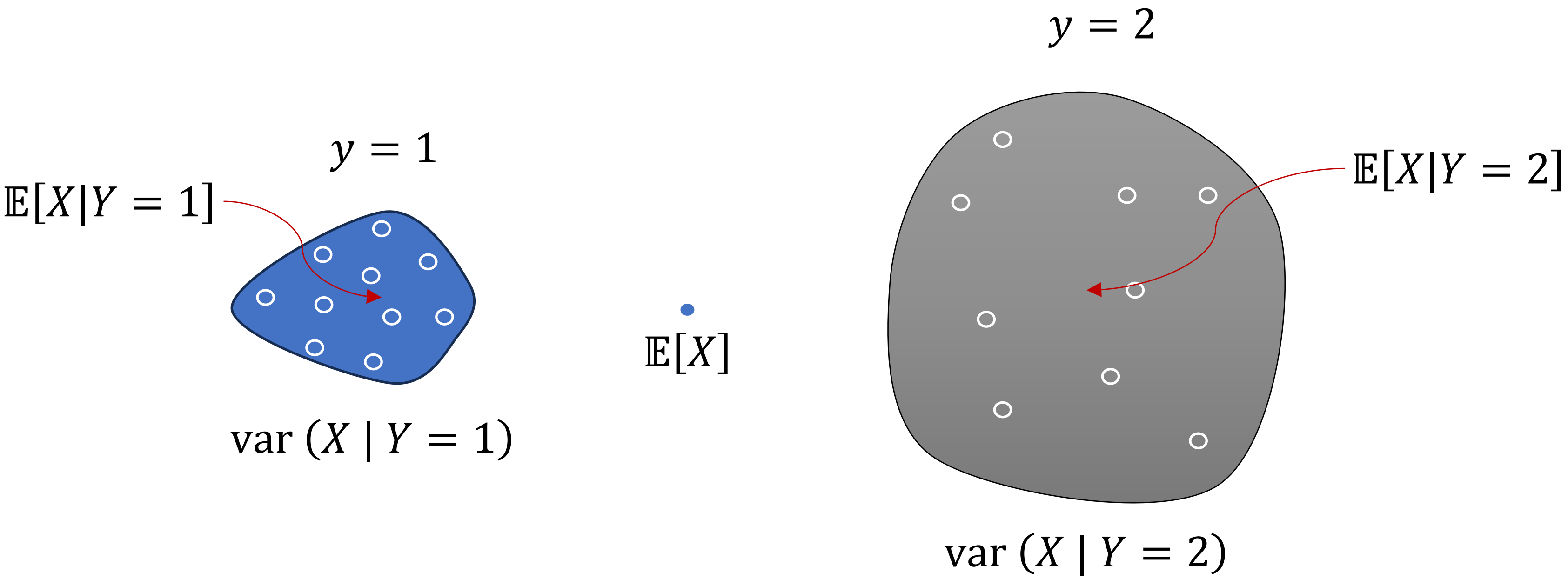
$$\text{var}\,(X \mid Y = y) = \mathbb{E}[(X - \mathbb{E}[X|Y])^2 \mid Y = y] \qquad \text{(conditional variance as number)}$$

$$\text{var}\,(X \mid Y) = \mathbb{E}[(X - \mathbb{E}[X|Y])^2] \qquad \text{(conditional variance as a \textbf{r.v.})}$$

- The law of total variance is given as

$$\text{var}\,(X) = \underbrace{\mathbb{E}[\text{var}\,(X \mid Y)]}_{\text{(variance within)}} + \underbrace{\text{var}\big[\mathbb{E}[X|Y]\big]}_{\text{(variance between)}}$$

Law of total variance



Law of total variance

Exercise 4 – Find the total variance of score in the previous question. Additional data

$y = 1$: $\frac{1}{10}\sum_{i=1}^{10}(x_i - 90)^2 = 10$ $\quad$ $y = 2$: $\frac{1}{20}\sum_{i=11}^{30}(x_i - 60)^2 = 20$

Sum of a random number of independent r.v.'s

- Consider the sum of a random number of independent randoms. How do we find expectation and variance of a such a sum? In practice, often we encounter this problem in insurance industry when both size and number of insurance claims are random. We also witness this in jump models where the frequency and size of the jump are random.

$$Y = X_1 + X_2 + \cdots + X_N \qquad X_i \text{ are independent r.v.'s, } N \text{ is a random integer}$$

$$\mathbb{E}[Y|N = n] = \mathbb{E}[X_1 + X_2 + \cdots + X_N \mid N = n] = \mathbb{E}[X_1 + X_2 + \cdots + X_n] = n\mathbb{E}[X]$$

$$\mathbb{E}[Y|N] = N\mathbb{E}[X]$$

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y|N]] = \mathbb{E}\big[N\mathbb{E}[X]\big] = \mathbb{E}[N]\mathbb{E}[X]$$

Sum of a random number of independent r.v.'s

- What about the variance of the sum. It can be calculated using law of total variance

$$\text{var}\,(Y) = \mathbb{E}[\text{var}\,(Y \mid N)] + \text{var}\big[\mathbb{E}[Y|N]\big]$$

$$\text{var}\,(Y \mid N = n) = \text{var}\,[X_1 + X_2 + \cdots + X_N \mid N = n] = \text{var}[X_1 + X_2 + \cdots + X_n] = n\mathbb{V}[X]$$

$$\text{var}\,(Y \mid N) = N\mathbb{V}[X]$$

$$\mathbb{E}[\text{var}\,(Y \mid N)] = \mathbb{E}[N]\mathbb{V}[X]$$

$$\mathbb{E}[Y|N] = N\mathbb{E}[X]$$

$$var\,(\mathbb{E}[Y|N]) = (\mathbb{E}[X])^2\mathbb{V}[N]$$

$$\text{var}\,(Y) = \mathbb{E}[N]\mathbb{V}[X] + (\mathbb{E}[X])^2\mathbb{V}[N]$$